

12

Parameterization of Stabilizing Controllers

The basic configuration of the feedback systems considered in this chapter is an LFT as shown in Figure 12.1, where G is the generalized plant with two sets of inputs: the exogenous inputs w , which include disturbances and commands, and control inputs u . The plant G also has two sets of outputs: the measured (or sensor) outputs y and the regulated outputs z . K is the controller to be designed. A control problem in this setup is either to analyze some specific properties, e.g., stability or performance, of the closed-loop or to design the feedback control K such that the closed-loop system is stable in some appropriate sense and the error signal z is specified, i.e., some performance condition is satisfied. In this chapter we are only concerned with the basic internal stabilization problems. We will see again that this setup is very convenient for other general control synthesis problems in the coming chapters.

Suppose that a given feedback system is feedback stabilizable. In this chapter, the problem we are mostly interested in is parameterizing all controllers that stabilize the system. The *parameterization* of all internally stabilizing controllers was first introduced by Youla et al [1976]; in their parameterization, the *coprime factorization* technique is used. All of the existing results are mainly in the frequency domain although they can also be transformed to state-space descriptions. In this chapter, we consider this issue in the general setting and directly in state space without adopting coprime factorization technique. The construction of the controller parameterization is done via considering a sequence of special problems, which are so-called *full information (FI)* problems, *disturbance feedforward (DF)* problems, *full control (FC)* problems and *output estimation*

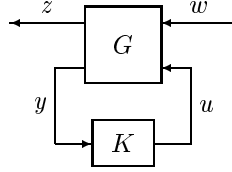


Figure 12.1: General System Interconnection

(*OE*) problems. On the other hand, these special problems are also of importance in their own right.

In addition to presenting the controller parameterization, this chapter also aims at introducing the synthesis machinery, which is essential in some control syntheses (the $\mathcal{H}_2$ and $\mathcal{H}_\infty$ control in Chapter 14 and Chapter 16), and at seeing how it works in the controller parameterization problem. The structure of this chapter is as follows: in section 12.1, the conditions for the existence of a stabilizing controller are examined. In section 12.2, we shall examine the stabilization of different special problems and establish the relations among them. In section 12.3, the construction of controller parameterization for the general output feedback problem will be considered via the special problems *FI*, *DF*, *FC* and *OE*. In section 12.4, the structure of the controller parameterization is displayed. Section 12.5 shows the closed-loop transfer matrix in terms of the parameterization. Section 12.6 considers an alternative approach to the controller parameterization using coprime factorizations and establishes the connections with the state space approach. This section can be either studied independently of all the preceding sections or skipped over without loss of continuity.

12.1 Existence of Stabilizing Controllers

Consider a system described by the standard block diagram in Figure 12.1. Assume that $G(s)$ has a *stabilizable and detectable* realization of the form

$$G(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right]. \quad (12.1)$$

The stabilization problem is to find feedback mapping K such that the closed-loop system is internally stable; the well-posedness is required for this interconnection. This general synthesis question will be called the *output feedback (OF)* problem.

Definition 12.1 A proper system G is said to be *stabilizable* through output feedback if there exists a proper controller K internally stabilizing G in Figure 12.1. Moreover, a proper controller $K(s)$ is said to be *admissible* if it internally stabilizes G .

The following result is standard and follows from Chapter 3.

Lemma 12.1 *There exists a proper K achieving internal stability iff (A, B_2) is stabilizable and (C_2, A) is detectable. Further, let F and L be such that $A + B_2F$ and $A + LC_2$ are stable, then an observer-based stabilizing controller is given by*

$$K(s) = \left[\frac{A + B_2F + LC_2 + LD_{22}F}{F} \mid \frac{-L}{0} \right].$$

Proof. $(\Leftarrow)$ By the stabilizability and detectability assumptions, there exist F and L such that $A + B_2F$ and $A + LC_2$ are stable. Now let $K(s)$ be the observer-based controller given in the lemma, then the closed-loop A -matrix is given by

$$\tilde{A} = \begin{bmatrix} A & B_2F \\ -LC_2 & A + B_2F + LC_2 \end{bmatrix}.$$

It is easy to check that this matrix is similar to the matrix

$$\begin{bmatrix} A + LC_2 & 0 \\ -LC_2 & A + B_2F \end{bmatrix}.$$

Thus the spectrum of $\tilde{A}$ equals the union of the spectra of $A + LC_2$ and $A + B_2F$. In particular, $\tilde{A}$ is stable.

$(\Rightarrow)$ If (A, B_2) is not stabilizable or if (C_2, A) is not detectable, then there are some eigenvalues of $\tilde{A}$ which are fixed in the right half-plane, no matter what the compensator is. The details are left as an exercise. $\square$

The stabilizability and detectability conditions of (A, B_2, C_2) are assumed throughout the remainder of this chapter¹. It follows that the realization for G_{22} is stabilizable and detectable, and these assumptions are enough to yield the following result:

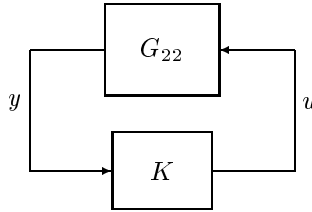


Figure 12.2: Equivalent Stabilization Diagram

¹It should be clear that the stabilizability and detectability of a realization for G do not guarantee the stabilizability and/or detectability of the corresponding realization for G_{22} .

Lemma 12.2 Suppose (A, B_2, C_2) is stabilizable and detectable and $G_{22} = \left[\begin{array}{c|c} A & B_2 \\ \hline C_2 & D_{22} \end{array} \right]$. Then the system in Figure 12.1 is internally stable iff the one in Figure 12.2 is internally stable.

In other words, $K(s)$ internally stabilizes $G(s)$ if and only if it internally stabilizes G_{22} .

Proof. The necessity follows from the definition. To show the sufficiency, it is sufficient to show that the system in Figure 12.1 and that in Figure 12.2 share the same A -matrix, which is obvious. $\square$

From Lemma 12.2, we see that the stabilizing controller for G depends only on G_{22} . Hence all stabilizing controllers for G can be obtained by using only G_{22} , which is how it is usually done in the conventional Youla parameterization. However, it will be shown that the general setup is very convenient and much more useful since any closed-loop system information can also be considered in the same framework.

Remark 12.1 There should be no confusion between a given realization for a transfer matrix G_{22} and the inherited realization from G where G_{22} is a submatrix. A given realization for G_{22} may be stabilizable and detectable while the inherited realization may be not. For instance,

$$G_{22} = \frac{1}{s+1} = \left[\begin{array}{c|c} -1 & 1 \\ \hline 1 & 0 \end{array} \right]$$

is a minimal realization but the inherited realization of G_{22} from

$$\left[\begin{array}{cc|cc} G_{11} & G_{12} & 0 & 1 \\ G_{21} & G_{22} & 0 & 0 \\ \hline & & 1 & 0 \end{array} \right] = \left[\begin{array}{cc|cc} -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ \hline 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{array} \right]$$

is

$$G_{22} = \left[\begin{array}{cc|c} -1 & 0 & 1 \\ 0 & 1 & 0 \\ \hline 1 & 0 & 0 \end{array} \right] \quad \left(= \frac{1}{s+1} \right)$$

which is neither stabilizable nor detectable. $\heartsuit$

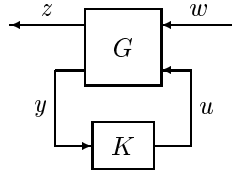
12.2 Duality and Special Problems

In this section, we will discuss four problems from which the output feedback solutions are constructed via a separation argument. These special problems are fundamental to the approach taken for synthesis in this book, and, as we shall see, they are also of importance in their own right.

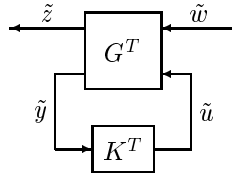
12.2.1 Algebraic Duality and Special Problems

Before we get into the study of the algebraic structure of control systems, we now introduce the concept of algebraic duality which will play an important role. It is well known that the concepts of controllability (stabilizability) and observability (detectability) of a system (C, A, B) are dual because of the duality between (C, A, B) and (B^T, A^T, C^T) . So, to deal with the issues related to a system's controllability and/or observability, we only need to examine the issues related to the observability and/or controllability of its dual system, respectively. The notion of duality can be generalized to a general setting.

Consider a standard system block diagram



where the plant G and controller K are assumed to be linear time invariant. Now consider another system shown below



whose plant and controller are obtained by transposing G and K . We can check easily that $T_{zw}^T = [\mathcal{F}_\ell(G, K)]^T = \mathcal{F}_\ell(G^T, K^T) = T_{\tilde{z}\tilde{w}}$. It is not difficult to see that K internally stabilizes G iff K^T internally stabilizes G^T . And we say that these two control structures are *algebraically dual*, especially, G^T and K^T which are dual objects of G and K , respectively. So as far as stabilization or other synthesis problems are concerned, we can obtain the results for G^T from the results for its dual object G if they are available.

Now, we consider some special problems which are related to the general OF problems stated in the last section and which are important in constructing the results for OF problems. The special problems considered here all pertain to the standard block diagram, but to different structures than G . The problems are labeled as

FI. Full information, with the corresponding plant

$$G_{FI} = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ \left[\begin{array}{c} I \\ 0 \end{array} \right] & \left[\begin{array}{c} 0 \\ I \end{array} \right] & \left[\begin{array}{c} 0 \\ 0 \end{array} \right] \end{array} \right].$$

FC. Full control, with the corresponding plant

$$G_{FC} = \left[\begin{array}{c|cc} A & B_1 & \left[\begin{array}{cc} I & 0 \end{array} \right] \\ \hline C_1 & D_{11} & \left[\begin{array}{cc} 0 & I \end{array} \right] \\ C_2 & D_{21} & \left[\begin{array}{cc} 0 & 0 \end{array} \right] \end{array} \right].$$

DF. Disturbance feedforward, with the corresponding plant

$$G_{DF} = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & I & 0 \end{array} \right].$$

OE. Output estimation, with the corresponding plant

$$G_{OE} = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & I \\ C_2 & D_{21} & 0 \end{array} \right].$$

The motivations for these special problems will be given later when they are considered. There are also two additional structures which are standard and which will not be considered in this chapter; they are

SF. State feedback

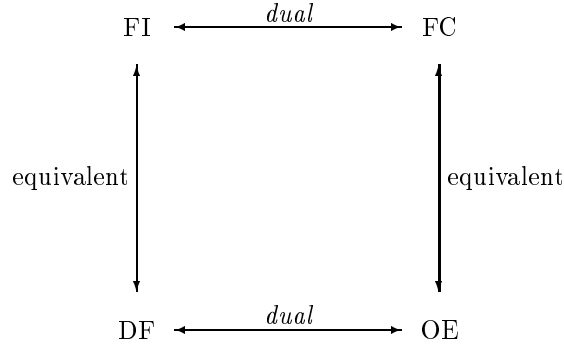
$$G_{SF} = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ I & 0 & 0 \end{array} \right].$$

OI. Output injection

$$G_{OI} = \left[\begin{array}{c|cc} A & B_1 & I \\ \hline C_1 & D_{11} & 0 \\ C_2 & D_{21} & 0 \end{array} \right].$$

Here we assume that all physical variables have compatible dimensions. We say that these special problems are special cases of *OF* problems in the sense that their structures are specified in comparison to *OF* problems.

The structure of transfer matrices shows clearly that FC, OE (and OI) are duals of FI, DF (and SF), respectively. These relationships are shown in the following diagram:



The precise meaning of “equivalent” in this diagram will be explained below.

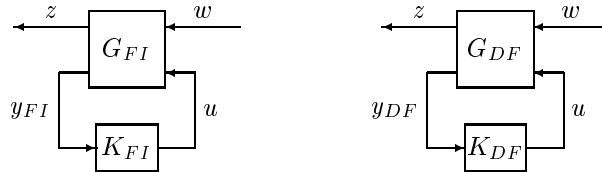
12.2.2 Full Information and Disturbance Feedforward

In the FI problem, the controller is provided with *Full Information* since $y = \begin{bmatrix} x \\ w \end{bmatrix}$.

For the *FI* problem, we only need to assume that (A, B_2) is stabilizable to guarantee the solvability. It is clear that if any output feedback control problem is to be solvable then the corresponding FI problem has to be solvable, provided *FI* is of the same structure with *OF* except for the specified parts.

To motivate the name *Disturbance Feedforward*, consider the special case with $C_2 = 0$. Then there is no feedback and the measurement is exactly w , where w is generally regarded as disturbance to the system. Only the disturbance, w , is fed through directly to the output. As we shall see, the feedback caused by $C_2 \neq 0$ does not affect the transfer function from w to the output z , but it does affect internal stability. In fact, the conditions for the solvability of the *DF* problem are that (A, B_2) is stabilizable and (C_2, A) is detectable.

Now we examine the connection between the DF problem and the FI problem and show the meaning of their equivalence. Suppose that we have controllers K_{FI} and K_{DF} and let T_{FI} and T_{DF} denote the closed-loop T_{zw} s in

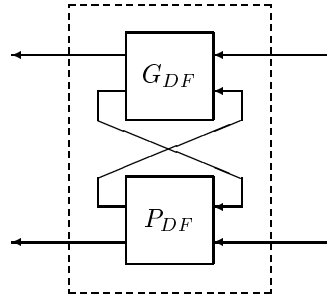


The question of interest is as follows: given either the K_{FI} or the K_{DF} controller, can we construct the other in such a way that $T_{FI} = T_{DF}$? The answer is positive. Actually, we have the following:

Lemma 12.3 *Let G_{FI} and G_{DF} be given as above. Then*

$$(i) \quad G_{DF}(s) = \begin{bmatrix} I & 0 & 0 \\ 0 & C_2 & I \end{bmatrix} G_{FI}(s).$$

(ii) $G_{FI} = \mathcal{S}(G_{DF}, P_{DF})$ (where $\mathcal{S}(\cdot, \cdot)$ denotes the star-product)



$$P_{DF}(s) = \left[\begin{array}{c|cc} A - B_1 C_2 & B_1 & B_2 \\ \hline 0 & 0 & I \\ \left[\begin{array}{c} I \\ -C_2 \end{array} \right] & \left[\begin{array}{c} 0 \\ I \end{array} \right] & \left[\begin{array}{c} 0 \\ 0 \end{array} \right] \end{array} \right].$$

Proof. Part (i) is obvious. Part (ii) follows from applying the star product formula. Nevertheless, we will give a direct proof to show the system structure. Let x and $\hat{x}$ denote the state of G_{DF} and P_{DF} , respectively. Take $e := x - \hat{x}$ and $\hat{x}$ as the states of the resulting interconnected system; then its realization is

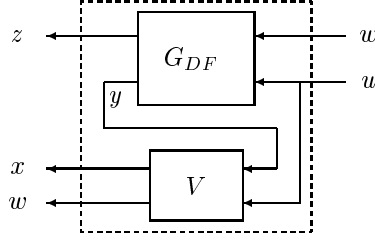
$$\left[\begin{array}{cc|cc} A - B_1 C_2 & 0 & 0 & 0 \\ B_1 C_2 & A & B_1 & B_2 \\ \hline C_1 & C_1 & D_{11} & D_{12} \\ \left[\begin{array}{c} 0 \\ C_2 \end{array} \right] & \left[\begin{array}{c} I \\ 0 \end{array} \right] & \left[\begin{array}{c} 0 \\ I \end{array} \right] & \left[\begin{array}{c} 0 \\ 0 \end{array} \right] \end{array} \right] = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ \left[\begin{array}{c} I \\ 0 \end{array} \right] & \left[\begin{array}{c} 0 \\ I \end{array} \right] & \left[\begin{array}{c} 0 \\ 0 \end{array} \right] \end{array} \right]$$

which is exactly G_{FI} , as claimed. $\square$

Remark 12.2 There is an alternative way to see part (ii). The fact is that in the DF problems, the disturbance w and the system states x can be solved in terms of y and u :

$$\begin{bmatrix} x \\ w \end{bmatrix} = \left[\begin{array}{c|cc} A - B_1 C_2 & B_1 & B_2 \\ \hline I & 0 & 0 \\ -C_2 & I & 0 \end{array} \right] \begin{bmatrix} y \\ u \end{bmatrix} =: V \begin{bmatrix} y \\ u \end{bmatrix}.$$

Now connect V up with G_{DF} as shown below



Then it is easy to show that the transfer function from (w, u) to (z, x, w) is G_{FI} , and, furthermore, that the internal stability is not changed if $A - B_1C_2$ is stable. $\heartsuit$

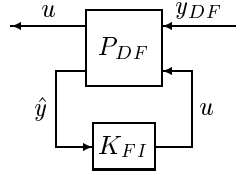
The following theorem follows immediately:

Theorem 12.4 *Let G_{FI} , G_{DF} , and P_{DF} be given as above.*

- (i) $K_{FI} := K_{DF} \begin{bmatrix} C_2 & I \end{bmatrix}$ internally stabilizes G_{FI} if K_{DF} internally stabilizes G_{DF} . Furthermore,

$$\mathcal{F}_l(G_{FI}, K_{DF} \begin{bmatrix} C_2 & I \end{bmatrix}) = \mathcal{F}_l(G_{DF}, K_{DF}).$$

- (ii) Suppose $A - B_1C_2$ is stable. Then $K_{DF} := \mathcal{F}_l(P_{DF}, K_{FI})$ as shown below



internally stabilizes G_{DF} if K_{FI} internally stabilizes G_{FI} . Furthermore,

$$\mathcal{F}_l(G_{DF}, \mathcal{F}_l(P_{DF}, K_{FI})) = \mathcal{F}_l(G_{FI}, K_{FI}).$$

Remark 12.3 This theorem shows that if $A - B_1C_2$ is stable, then problems FI and DF are equivalent in the above sense. Note that the transfer function from w to y_{DF} is

$$G_{21}(s) = \left[\begin{array}{c|c} A & B_1 \\ \hline C_2 & I \end{array} \right].$$

Hence this stability condition implies that $G_{21}(s)$ has neither right half plane invariant zeros nor hidden unstable modes. $\heartsuit$

12.2.3 Full Control and Output Estimation

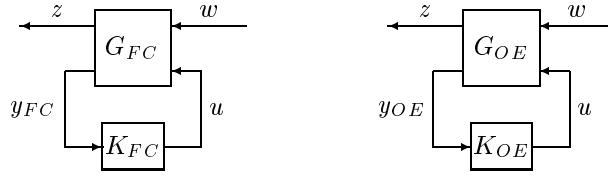
For the FC problem, the term *Full Control* is used because the controller has full access to both the state through output injection and the output z . The only restriction on the controller is that it must work with the measurement y . This problem is dual to the FI case and has the dual solvability condition to the FI problem, which is also guaranteed by the assumptions on *OF* problems. The solutions to this kind of control problem can be obtained by first transposing G_{FC} , and solving the corresponding *FI* problem, and then transposing back.

On the other hand, problem *OE* is dual to *DF*. Thus the discussion of the *DF* problem is relevant here, when appropriately dualized. And the solvability conditions for the *OE* problem are that (A, B_2) is stabilizable and (C_2, A) is detectable. To examine the physical meaning of *output estimation*, first note that

$$z = C_1 x + D_{11} w + u$$

where z is to be controlled by an appropriately designed control u . In general, our control objective will be to specify z in some well-defined mathematical sense. To put it in other words, it is desired to find a u that will estimate $C_1 x + D_{11} w$ in such defined mathematical sense. So this kind of control problem can be regarded as an estimation problem. We are focusing on this particular estimation problem because it is the one that arises in solving the output feedback problem. A more conventional estimation problem would be the special case where no internal stability condition is imposed and $B_2 = 0$. Then the problem would be that of estimating the output z given the measurement y . This special case motivates the term *output estimation* and can be obtained immediately from the results obtained for the general case.

The following discussion will explain the meaning of equivalence between *FC* and *OE* problems. Consider the following *FC* and *OE* diagrams:



We have similar results to the ones in the last subsection:

Lemma 12.5 *Let G_{FC} and G_{OE} be given as above. Then*

$$(i) \quad G_{OE}(s) = G_{FC}(s) \begin{bmatrix} I & 0 \\ 0 & B_2 \\ 0 & I \end{bmatrix}$$

(ii) $G_{FC} = S(G_{OE}, P_{OE})$, where P_{OE} is given by

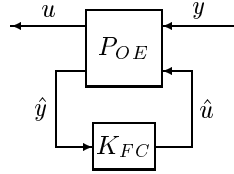
$$P_{OE}(s) = \left[\begin{array}{c|c} A - B_2 C_1 & 0 \\ \hline C_1 & 0 \\ C_2 & I \end{array} \left| \begin{array}{c} I \quad -B_2 \\ 0 \quad I \\ 0 \quad 0 \end{array} \right. \right].$$

Theorem 12.6 Let G_{FC} , G_{OE} , and P_{OE} be given as above.

(i) $K_{FC} := \begin{bmatrix} B_2 \\ I \end{bmatrix} K_{OE}$ internally stabilizes G_{FC} if K_{OE} internally stabilizes G_{OE} .
Furthermore,

$$\mathcal{F}_\ell(G_{FC}, \begin{bmatrix} B_2 \\ I \end{bmatrix} K_{OE}) = \mathcal{F}_\ell(G_{OE}, K_{OE}).$$

(ii) Suppose $A - B_2 C_1$ is stable. Then $K_{OE} := \mathcal{F}_\ell(P_{OE}, K_{FC})$, as shown below



internally stabilizes G_{OE} if K_{FC} internally stabilizes G_{FC} . Furthermore,

$$\mathcal{F}_\ell(G_{OE}, \mathcal{F}_\ell(P_{OE}, K_{FC})) = \mathcal{F}_\ell(G_{FC}, K_{FC}).$$

Remark 12.4 It is seen that if $A - B_2 C_1$ is stable, then FC and OE problems are equivalent in the above sense. This condition implies that the transfer matrix $G_{12}(s)$ from u to z has neither right half-plane invariant zeros nor hidden unstable modes, which indicates that it has a stable inverse. $\heartsuit$

12.3 Parameterization of All Stabilizing Controllers

12.3.1 Problem Statement and Solution

Consider again the standard system block diagram in Figure 12.1 with

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right] = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix}.$$

Suppose (A, B_2) is stabilizable and (C_2, A) is detectable. In this section we discuss the following problem:

Given a plant G , parameterize all controllers K that internally stabilize G .

This parameterization for all stabilizing controllers is usually called Youla parameterization. As we have mentioned early, the stabilizing controllers for G will depend only on G_{22} . However, it is more convenient to consider the problem in the general framework as will be shown. The parameterization of all stabilizing controllers is easy when the plant itself is stable.

Theorem 12.7 *Suppose $G \in \mathcal{RH}_\infty$; then the set of all stabilizing controllers can be described as*

$$K = Q(I + G_{22}Q)^{-1} \quad (12.2)$$

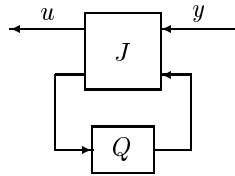
for any $Q \in \mathcal{RH}_\infty$ and $I + D_{22}Q(\infty)$ nonsingular.

Remark 12.5 This result is very natural considering Corollary 5.5, which says that a controller K stabilizes a stable plant G_{22} iff $K(I - G_{22}K)^{-1}$ is stable. Now suppose $Q = K(I - G_{22}K)^{-1}$ is a stable transfer matrix, then K can be solved from this equation which gives exactly the controller parameterization in the above theorem. $\heartsuit$

Proof. Note that $G_{22}(s)$ is stable by the assumptions on G . Now use straightforward algebra to verify that the controllers given above stabilize G_{22} . On the other hand, suppose K_0 is a stabilizing controller; then $Q_0 := K_0(I - G_{22}K_0)^{-1} \in \mathcal{RH}_\infty$, so K_0 can be expressed as $K_0 = Q_0(I + G_{22}Q_0)^{-1}$. Note that the invertibility in the last equation is guaranteed by the well posedness of the interconnected system with controller K_0 since $I + D_{22}Q_0(\infty) = (I - D_{22}K_0(\infty))^{-1}$. $\square$

However, if G is not stable, the parameterization is much more complicated. The results can be more conveniently stated using state space representations.

Theorem 12.8 *Let F and L be such that $A + LC_2$ and $A + B_2F$ are stable, and then all controllers that internally stabilize G can be parameterized as the transfer matrix from y to u below*



$$J = \left[\begin{array}{c|cc} A + B_2F + LC_2 + LD_{22}F & -L & B_2 + LD_{22} \\ \hline F & 0 & I \\ \hline -(C_2 + D_{22}F) & I & -D_{22} \end{array} \right]$$

with any $Q \in \mathcal{RH}_\infty$ and $I + D_{22}Q(\infty)$ nonsingular.

A non-constructive proof of the theorem can be given by using the same argument as in the proof of Theorem 12.7, i.e., first verify that any controller given by the formula indeed stabilizes the system G , and then show that we can construct a stable Q for any

given stabilizing controller K . This approach, however, does not give much insight into the controller construction and thus can not be generalized to other synthesis problems.

The conventional Youla approach to this problem is via coprime factorization [Youla et al, 1976, Vidyasagar, 1985, Desoer et al, 1982], which will be adopted in the later part of this chapter as an alternative approach.

In the following sections, we will present a novel approach to this problem without adopting coprime factorizations. The idea of this approach is to reduce the output feedback problem into some simpler problems, such as FI and OE or FC and DF which admit simple solutions, and then to solve the output feedback problem by the separation argument. The advantages of this approach are that it is simple and that many other synthesis problems, such as $\mathcal{H}_2$ and $\mathcal{H}_\infty$ optimal control problems in Chapters 14 and 16, can be solved by using the same machinery.

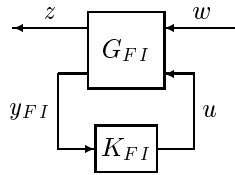
Readers should bear in mind that our objective here is to find all admissible controllers for the OF problem. So at first, we will try to build up enough tools for this objective by considering the special problems. We will see that it is not necessary to parameterize all stabilizing controllers for these special problems to get the required tools. Instead, we only parameterize some equivalent classes of controllers which generate the *same control action*.

Definition 12.2 Two controllers K and $\hat{K}$ are of *equivalent control actions* if their corresponding closed loop transfer matrices are identical, i.e. $\mathcal{F}_l(G, K) = \mathcal{F}_l(G, \hat{K})$, written as $K \cong \hat{K}$.

Algebraically, the controller equivalence is an *equivalence relation*. We will see that for different special problems we have different refined versions of this relation. We will also see that the characterizations of equivalent classes of stabilizing controllers for special problems are good enough to construct the parameterization of all stabilizing controllers for the OF problem. In the next two subsections, we mainly consider the stabilizing controller characterizations for special problems. Also, we use the solutions to these special problems and the approach provided in the last section to characterize all stabilizing controllers of OF problems.

12.3.2 Stabilizing Controllers for FI and FC Problems

In this subsection, we first examine the FI structure



where the transfer matrix G_{FI} is given in section 12.2. The purpose of this subsection is to characterize the equivalent classes of stabilizing controllers K_{FI} that stabilize

internally G_{FI} and to build up enough tools to be used later. For this problem, we say *two controllers K_{FI} and $\tilde{K}_{FI}$ are equivalent if they produce the same closed-loop transfer function from w to u* . Obviously, this also guarantees that $\mathcal{F}_l(G_{FI}, K_{FI}) = \mathcal{F}_l(G_{FI}, \tilde{K}_{FI})$. The characterization of equivalent classes of controllers can be suitably called the *control parameterization* in contrast with controller parameterization. Note that the same situation will occur in *FC* problems by duality.

Since we have full information for feedback, our controller will have the following general form:

$$K_{FI} = \begin{bmatrix} K_1(s) & K_2(s) \end{bmatrix}$$

with $K_1(s)$ stabilizing internally $(sI - A)^{-1}B_2$ and arbitrary $K_2(s) \in \mathcal{RH}_\infty$. Note that the stability of K_2 is required to guarantee the internal stability of the whole system since w is fed directly through K_2 .

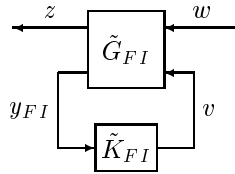
Lemma 12.9 *Let F be a constant matrix such that $A + B_2F$ is stable. Then all stabilizing controllers, in the sense of generating all admissible closed-loop transfer functions, for *FI* can be parameterized as*

$$K_{FI} \cong \begin{bmatrix} F & Q \end{bmatrix}$$

with any $Q \in \mathcal{RH}_\infty$.

Note that for the parameter matrix $Q \in \mathcal{RH}_\infty$, it is reasonable to assume that *the realization of $Q(s)$ is stabilizable and detectable*.

Proof. It is easy to see that the controller given in the above formula stabilizes the system G_{FI} . Hence we only need to show that the given set of controllers parameterizes all stabilizing control action, u , i.e., there is a choice of $Q \in \mathcal{RH}_\infty$ such that the transfer functions from w to u for any stabilizing controller $K_{FI} = \begin{bmatrix} K_1(s) & K_2(s) \end{bmatrix}$ and for $K_{FI}^0 = \begin{bmatrix} F & Q \end{bmatrix}$ are the same. To show this, make a change of control variable as $v = u - Fx$, where x denotes the state of the system G_{FI} ; then the system with the controller K_{FI} will be as shown in the following diagram:



with

$$\tilde{G}_{FI} = \left[\begin{array}{c|cc} A + B_2F & B_1 & B_2 \\ \hline C_1 + D_{12}F & D_{11} & D_{12} \\ \hline \begin{bmatrix} I \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ I \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{array} \right], \quad \tilde{K}_{FI} = K_{FI} - [F \ 0].$$

Let Q be the transfer matrix from w to v ; Q belongs to $\mathcal{RH}_\infty$ by internal stability. Then $u = Fx + v = Fx + Qw$, so $K_{FI} \cong \begin{bmatrix} F & Q \end{bmatrix}$. $\square$

Remark 12.6 The equivalence of $K_{FI} \cong K_{FI}^0$ in the above equation can actually be shown by directly computing Q from equating the transfer matrices from w to u for the cases of K_{FI} and K_{FI}^0 . In fact, the transfer matrices from w to u with K_{FI} and K_{FI}^0 are given by

$$\left[I - K_1(sI - A)^{-1}B_2 \right]^{-1} K_1(sI - A)^{-1}B_1 + \left[I - K_1(sI - A)^{-1}B_2 \right]^{-1} K_2 \quad (12.3)$$

and

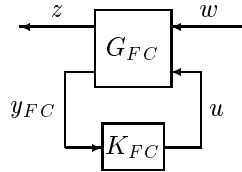
$$\left[I - F(sI - A)^{-1}B_2 \right]^{-1} F(sI - A)^{-1}B_1 + \left[I - F(sI - A)^{-1}B_2 \right]^{-1} Q, \quad (12.4)$$

respectively. We can verify that

$$Q = K_2 + (K_1 - F)(sI - A - B_2K_1)^{-1}(B_2K_2 + B_1)$$

is stable and makes the formulas in (12.3) and (12.4) equal. $\heartsuit$

Now we consider the dual FC problem; the system diagram pertinent to this case is



Dually, we say controllers K_{FC} and $\hat{K}_{FC}$ are equivalent in the sense that the same injection inputs y_{FC} 's produce the same outputs z 's. This also guarantees the identity of their resulting closed-loop transfer matrices from w to z . And we also have

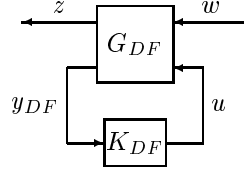
Lemma 12.10 *Let L be a constant matrix such that $A + LC_2$ is stable. Then the set of equivalent classes of all stabilizing controllers for FC in the above sense can be parameterized as*

$$K_{FC} \cong \begin{bmatrix} L \\ Q \end{bmatrix}$$

with any $Q \in \mathcal{RH}_\infty$.

12.3.3 Stabilizing Controllers for DF and OE Problems

In the DF case we have the following system diagram:



The transfer matrix is given as in section 12.2.1. We will further assume that $A - B_1C_2$ is stable in this subsection. It should be pointed out that the existence of a stabilizing controller for this system is guaranteed by the stabilizability of (A, B_2) and detectability of (C_2, A) . Hence this assumption is not necessary for our problem to be solvable; however, it does simplify the solution.

We will now parameterize stabilizing controllers for G_{DF} by invoking the relationship between the FI problem and DF problem established in section 12.2. We say that *the controllers K_{DF} and $\hat{K}_{DF}$ are equivalent for the DF problem if the two transfer matrices from w to u in the above diagram are the same*. Of course, the resulting two closed-loop transfer matrices from w to z are identical.

Remark 12.7 By the equivalence between FI and DF problems, it is easy to show that if $K_{DF} \cong \hat{K}_{DF}$ in the DF structure, then $K_{DF} \begin{bmatrix} C_2 & I \end{bmatrix} \cong \hat{K}_{DF} \begin{bmatrix} C_2 & I \end{bmatrix}$ in the corresponding FI structure. We also have that if $K_{FI} \cong \hat{K}_{FI}$, then $\mathcal{F}_l(P_{DF}, K_{FI}) \cong \mathcal{F}_l(P_{DF}, \hat{K}_{FI})$. $\heartsuit$

Now we construct the parameterization of equivalent classes of stabilizing controllers in DF structure via the tool we have developed in section 12.2.

Let $K_{DF}(s)$ be a stabilizing control for DF ; then $K_{FI}(s) = K_{DF}(s) \begin{bmatrix} C_2 & I \end{bmatrix}$ stabilizes the corresponding G_{FI} . Assume $K_{FI} \cong \hat{K}_{FI} = \begin{bmatrix} F & Q \end{bmatrix}$ for some $Q \in \mathcal{RH}_\infty$; then $\hat{K}_{FI}$ stabilizes G_{FI} and $\mathcal{F}_l(J_{DF}, Q) = \mathcal{F}_l(P_{DF}, \hat{K}_{FI})$ where

$$J_{DF} = \left[\begin{array}{c|cc} A + B_2F - B_1C_2 & B_1 & B_2 \\ \hline F & 0 & I \\ -C_2 & I & 0 \end{array} \right]$$

with F such that $A + B_2F$ is stable. Hence by Theorem 12.4, $\hat{K}_{DF} := \mathcal{F}_l(J_{DF}, Q)$ stabilizes G_{DF} for any $Q \in \mathcal{RH}_\infty$. Since $K_{FI} \cong \hat{K}_{FI}$, by Remarks 12.7 we have $K_{DF} \cong \hat{K}_{DF} = \mathcal{F}_l(J_{DF}, Q)$. This equation characterizes an equivalent class of all controllers for the DF problem by the equivalence of FI and DF .

In fact, we have the following lemma which shows that the above construction of parameterization characterizes all stabilizing controllers for the DF problem.

Lemma 12.11 *All stabilizing controllers for the DF problem can be characterized by $K_{DF} = \mathcal{F}_\ell(J_{DF}, Q)$ with $Q \in \mathcal{RH}_\infty$, where J_{DF} is given as above.*

Proof. We have already shown that the controller $K_{DF} = \mathcal{F}_\ell(J_{DF}, Q)$ for any given $Q \in \mathcal{RH}_\infty$ does internally stabilize G_{DF} . Now let K_{DF} be any stabilizing controller for G_{DF} ; then $\mathcal{F}_\ell(\hat{J}_{DF}, K_{DF}) \in \mathcal{RH}_\infty$ where

$$\hat{J}_{DF} = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline -F & 0 & I \\ C_2 & I & 0 \end{array} \right].$$

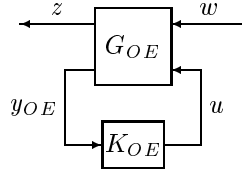
($\hat{J}_{DF}$ is stabilized by K_{DF} since it has the same ' G_{22} ' matrix as G_{DF} .)

Let $Q_0 := \mathcal{F}_\ell(\hat{J}_{DF}, K_{DF}) \in \mathcal{RH}_\infty$; then $\mathcal{F}_\ell(J_{DF}, Q_0) = \mathcal{F}_\ell(J_{DF}, \mathcal{F}_\ell(\hat{J}_{DF}, K_{DF})) =: \mathcal{F}_\ell(J_{tmp}, K_{DF})$, where J_{tmp} can be obtained by using the state space star product formula given in Chapter 10:

$$\begin{aligned} J_{tmp} &= \left[\begin{array}{cc|cc} A - B_1C_2 + B_2F & -B_2F & B_1 & B_2 \\ -B_1C_2 & A & B_1 & B_2 \\ \hline F & -F & 0 & I \\ -C_2 & C_2 & I & 0 \end{array} \right] \\ &= \left[\begin{array}{cc|cc} A - B_1C_2 & -B_2F & B_1 & B_2 \\ 0 & A + B_2F & 0 & 0 \\ \hline 0 & -F & 0 & I \\ 0 & C_2 & I & 0 \end{array} \right] \\ &= \left[\begin{array}{cc} 0 & I \\ I & 0 \end{array} \right]. \end{aligned}$$

Hence $\mathcal{F}_\ell(J_{DF}, Q_0) = \mathcal{F}_\ell(J_{tmp}, K_{DF}) = K_{DF}$. This shows that any stabilizing controller can be expressed in the form of $\mathcal{F}_\ell(J_{DF}, Q_0)$ for some $Q_0 \in \mathcal{RH}_\infty$. $\square$

Now we turn to the dual *OE* case. The corresponding system diagram is shown as below:



We will assume that $A - B_2C_1$ is stable. Again this assumption is made only for the simplicity of the solution, it is not necessary for the stabilization problem to be

solvable. The parameterization of an equivalent class of stabilizing controllers for OE can be obtained by invoking Theorem 12.6 and the equivalent classes of controllers for FC . Here we say *controllers* K_{OE} and $\hat{K}_{OE}$ are *equivalent* if the transfer matrices from y_{OE} to z are the same. This also guarantees that the resulting closed-loop transfer matrices are identical.

Now we construct the parameterization for the OE structure as a dual case to DF . Assume that K_{OE} is an admissible controller for OE ; then $K_{FC} = \begin{bmatrix} B_2 \\ I \end{bmatrix} K_{OE} \cong \begin{bmatrix} L \\ Q \end{bmatrix} =: \hat{K}_{FC}$ for some $Q \in \mathcal{RH}_\infty$, and $\hat{K}_{FC}$ stabilizes G_{FC} and $\mathcal{F}_\ell(J_{OE}, Q) = \mathcal{F}_\ell(P_{OE}, \hat{K}_{FC})$, where

$$J_{OE} = \left[\begin{array}{c|cc} A - B_2 C_1 + L C_2 & L & -B_2 \\ \hline C_1 & 0 & I \\ C_2 & I & 0 \end{array} \right]$$

with L such that $A + L C_2$ is stable. Hence by Theorem 12.6, $\hat{K}_{OE} = \mathcal{F}_\ell(J_{OE}, Q)$ stabilizes G_{OE} for any $Q \in \mathcal{RH}_\infty$. Since $K_{OE} \cong \hat{K}_{OE}$, $K_{OE} \cong \mathcal{F}_\ell(J_{OE}, Q)$. In fact, we have the following lemma.

Lemma 12.12 *All admissible controllers for the OE problem can be characterized as $\mathcal{F}_\ell(J_{OE}, Q_0)$ with any $Q_0 \in \mathcal{RH}_\infty$, where J_{OE} is defined as above.*

Proof. The controllers in the form as stated in the theorem are admissible since the corresponding FC controllers internally stabilize resulting G_{FC} .

Now assume K_{OE} is any stabilizing controller for G_{OE} ; then $\mathcal{F}_\ell(\hat{J}_{OE}, K_{OE}) \in \mathcal{RH}_\infty$ where

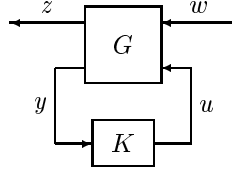
$$\hat{J}_{OE} = \left[\begin{array}{c|cc} A & -L & B_2 \\ \hline C_1 & 0 & I \\ C_2 & I & 0 \end{array} \right].$$

Let $Q_0 := \mathcal{F}_\ell(\hat{J}_{OE}, K_{OE}) \in \mathcal{RH}_\infty$. Then $\mathcal{F}_\ell(J_{OE}, Q_0) = \mathcal{F}_\ell(J_{OE}, \mathcal{F}_\ell(\hat{J}_{OE}, K_{OE})) = K_{OE}$, by using again the state space star product formula given in Chapter 10. This shows that any stabilizing controller can be expressed in the form of $\mathcal{F}_\ell(J_{OE}, Q_0)$ for some $Q_0 \in \mathcal{RH}_\infty$. $\square$

12.3.4 Output Feedback and Separation

We are now ready to give a complete proof for Theorem 12.8. We will assume the results of the special problems and show how to construct all admissible controllers for the OF problem from them. And we can also observe the separation argument as the

byproduct; this essentially involves reducing the *OF* problem to the combination of the simpler *FI* and *FC* problems. Moreover, we can see from the construction why the stability conditions of $A - B_1C_2$ and $A - B_2C_1$ in *DF* and *OE* problems were reasonably assumed and are automatically guaranteed in this case. Again we assume that the system has the following standard system block diagram:



with

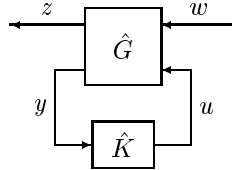
$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right]$$

and that (A, B_2) is stabilizable and (C_2, A) is detectable.

Proof of Theorem 12.8. Without loss of generality, we shall assume $D_{22} = 0$. For more general cases, i.e. $D_{22} \neq 0$, the mapping

$$\hat{K}(s) = K(s)(I - D_{22}K(s))^{-1}$$

is well defined if the closed-loop system is assumed to be well posed. Then the system in terms of $\hat{K}$ has the structure



where

$$\hat{G}(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & 0 \end{array} \right].$$

Now we construct the controllers for the *OF* problem with $D_{22} = 0$. Denote x the state of system G ; then the open-loop system can be written as

$$\begin{bmatrix} \dot{x} \\ z \\ y \end{bmatrix} = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & 0 \end{bmatrix} \begin{bmatrix} x \\ w \\ u \end{bmatrix}.$$

Since (A, B_2) is stabilizable, there is a constant matrix F such that $A + B_2F$ is stable. Note that $\begin{bmatrix} F & 0 \end{bmatrix}$ is actually a special FI stabilizing controller. Now let

$$v = u - Fx.$$

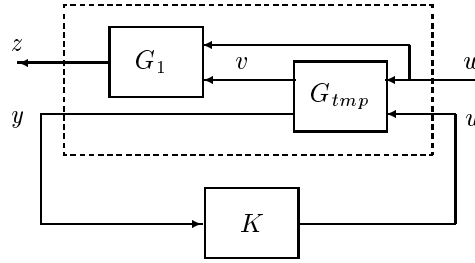
Then the system can be broken into two subsystems:

$$\begin{bmatrix} \dot{x} \\ z \end{bmatrix} = \begin{bmatrix} A + B_2F & B_1 & B_2 \\ C_1 + D_{12}F & D_{11} & D_{12} \end{bmatrix} \begin{bmatrix} x \\ w \\ v \end{bmatrix}$$

and

$$\begin{bmatrix} \dot{x} \\ v \\ y \end{bmatrix} = \begin{bmatrix} A & B_1 & B_2 \\ -F & 0 & I \\ C_2 & D_{21} & 0 \end{bmatrix} \begin{bmatrix} x \\ w \\ u \end{bmatrix}.$$

This can be shown pictorially below:



with

$$G_1 = \left[\begin{array}{c|cc} A + B_2F & B_1 & B_2 \\ \hline C_1 + D_{12}F & D_{11} & D_{12} \end{array} \right] \in \mathcal{RH}_\infty$$

and

$$G_{tmp} = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline -F & 0 & I \\ C_2 & D_{21} & 0 \end{array} \right].$$

Obviously, K stabilizes G if and only if K stabilizes G_{tmp} ; however, G_{tmp} is of OE structure. Now let L be such that $A + LC_2$ is stable. Then by Lemma 12.12 all controllers stabilizing G_{tmp} are given by

$$K = \mathcal{F}_\ell(J, Q)$$

where

$$J = \left[\begin{array}{c|cc} A + B_2F + LC_2 & L & -B_2 \\ \hline -F & 0 & I \\ C_2 & I & 0 \end{array} \right] = \left[\begin{array}{c|cc} A + B_2F + LC_2 & -L & B_2 \\ \hline F & 0 & I \\ -C_2 & I & 0 \end{array} \right].$$

This concludes the proof. $\square$

Remark 12.8 We can also get the same result by applying the dual procedure to the above construction, i.e., first use an output injection to reduce the OF problem to a DF problem. The separation argument is obvious since the synthesis of the OF problem can be reduced to FI and FC problems, i.e. the latter two problems can be designed independently. $\heartsuit$

Remark 12.9 Theorem 12.8 shows that any stabilizing controller $K(s)$ can be characterized as an LFT of a parameter matrix $Q \in \mathcal{RH}_\infty$, i.e., $K(s) = \mathcal{F}_\ell(J, Q)$. Moreover, using the same argument as in the proof of Lemma 12.11, a realization of $Q(s)$ in terms of K can be obtained as

$$Q := \mathcal{F}_\ell(\hat{J}, K)$$

where

$$\hat{J} = \left[\begin{array}{c|cc} A & -L & B_2 \\ \hline -F & 0 & I \\ C_2 & I & D_{22} \end{array} \right]$$

and where $K(s)$ has the stabilizable and detectable realization. $\heartsuit$

Now we can reconsider the characterizations of all stabilizing controllers for the special problems with some reasonable assumptions, i.e. the stability conditions of $A - B_1C_2$ and $A - B_2C_1$ for DF and OE problems which were assumed in the last section can be dropped.

If we specify

$$C_2 = \begin{bmatrix} I \\ 0 \end{bmatrix} \quad D_{21} = \begin{bmatrix} 0 \\ I \end{bmatrix} \quad D_{22} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

the OF problem, in its general setting, becomes the FI problem. We know that the solvability conditions for the FI problem are reduced, because of its special structure, to (A, B_2) as stabilizable. By assuming this, we can get the following result from the OF problem.

Corollary 12.13 *Let L_1 and F be such that $A + L_1$ and $A + B_2F$ are stable; then all controllers that stabilize G_{FI} can be characterized as $\mathcal{F}_\ell(J_{FI}, Q)$ with any $Q \in \mathcal{RH}_\infty$,*

where

$$J_{FI} = \left[\begin{array}{c|cc} A + B_2F + L_1 & L & -B_2 \\ \hline -F & 0 & I \\ \begin{bmatrix} I \\ 0 \end{bmatrix} & I & 0 \end{array} \right]$$

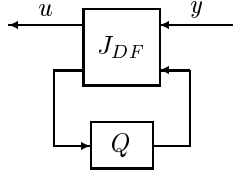
and where $L = (L_1 \ L_2)$ is the injection matrix for any L_2 with compatible dimensions.

In the same way, we can consider the FC problem as the special OF problem by specifying

$$B_2 = \begin{bmatrix} I & 0 \end{bmatrix} \quad D_{12} = \begin{bmatrix} 0 & I \end{bmatrix} \quad D_{22} = \begin{bmatrix} 0 & 0 \end{bmatrix}.$$

The $DF(OE)$ problem can also be considered as the special case of OF by simply setting $D_{21} = I$ ($D_{12} = I$) and $D_{22} = 0$.

Corollary 12.14 *Consider the DF problem, and assume that (C_2, A, B_2) is stabilizable and detectable. Let F and L be such that $A + LC_2$ and $A + B_2F$ are stable, and then all controllers that internally stabilize G can be parameterized as $\mathcal{F}_l(J_{DF}, Q)$ for some $Q \in \mathcal{RH}_\infty$, i.e. the transfer function from y to u is shown as below*



$$J_{DF} = \left[\begin{array}{c|cc} A + B_2F + LC_2 & -L & B_2 \\ \hline F & 0 & I \\ -C_2 & I & 0 \end{array} \right].$$

Remark 12.10 It would be interesting to compare this result with Lemma 12.11. It can be seen that Lemma 12.11 is a special case of this corollary. The condition that $A - B_1C_2$ is stable, which is required in Lemma 12.11, provides the natural injection matrix $L = -B_1$ which satisfies a partial condition in this corollary. ♡

12.4 Structure of Controller Parameterization

Let us recap what we have done. We begin with a stabilizable and detectable realization of G_{22}

$$G_{22} = \left[\begin{array}{c|c} A & B_2 \\ \hline C_2 & D_{22} \end{array} \right].$$

We choose F and L so that $A + B_2F$ and $A + LC_2$ are stable. Define J by the formula in Theorem 12.8. Then the proper K 's achieving internal stability are precisely those representable in Figure 12.3 and $K = \mathcal{F}_l(J, Q)$ where $Q \in \mathcal{RH}_\infty$ and $I + D_{22}Q(\infty)$ is invertible.

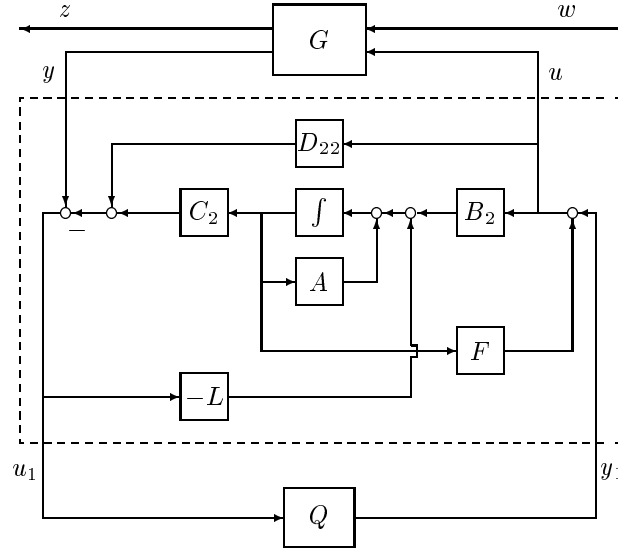


Figure 12.3: Structure of Stabilizing Controllers

It is interesting to note that the system in the dashed box is an observer-based stabilizing controller for G (or G_{22}). Furthermore, it is easy to show that the transfer function between (y, y_1) and (u, u_1) is J , i.e.,

$$\begin{bmatrix} u \\ u_1 \end{bmatrix} = J \begin{bmatrix} y \\ y_1 \end{bmatrix}.$$

It is also easy to show that the transfer matrix from y_1 to u_1 is zero.

This diagram of all stabilizing controller parameterization also suggests an interesting interpretation: every internal stabilization amounts to adding stable dynamics to the plant and then stabilizing the extended plant by means of an observer. The precise statement is as follows: for simplicity of the formulas, only the cases of strictly proper G_{22} and K are treated.

Theorem 12.15 *Assume that G_{22} and K are strictly proper and the system is Figure 12.1 is internally stable. Then G_{22} can be embedded in a system*

$$\left[\begin{array}{c|c} A_e & B_e \\ \hline C_e & 0 \end{array} \right]$$

where

$$A_e = \begin{bmatrix} A & 0 \\ 0 & A_a \end{bmatrix}, \quad B_e = \begin{bmatrix} B_2 \\ 0 \end{bmatrix}, \quad C_e = \begin{bmatrix} C_2 & 0 \end{bmatrix} \quad (12.5)$$

and where A_a is stable, such that K has the form

$$K = \left[\begin{array}{c|c} \frac{A_e + B_e F_e + L_e C_e}{F_e} & -L_e \\ \hline & 0 \end{array} \right] \quad (12.6)$$

where $A_e + B_e F_e$ and $A_e + L_e C_e$ are stable.

Proof. K is representable as in Figure 12.3 for some Q in $\mathcal{RH}_\infty$. For K to be strictly proper, Q must be strictly proper. Take a minimal realization of Q :

$$Q = \left[\begin{array}{c|c} A_a & B_a \\ \hline C_a & 0 \end{array} \right].$$

Since $Q \in \mathcal{RH}_\infty$, A_a is stable. Let x and x_a denote state vectors for J and Q , respectively, and write the equations for the system in Figure 12.3:

$$\begin{aligned} \dot{x} &= (A + B_2 F + L C_2)x - L y + B_2 y_1 \\ u &= F x + y_1 \\ u_1 &= -C_2 x + y \\ \dot{x}_a &= A_a x_a + B_a u_1 \\ y_1 &= C_a x_a \end{aligned}$$

These equations yield

$$\begin{aligned} \dot{x}_e &= (A_e + B_e F_e + L_e C_e)x_e - L_e y \\ u &= F_e x_e \end{aligned}$$

where

$$x_e := \begin{bmatrix} x \\ x_a \end{bmatrix}, \quad F_e := \begin{bmatrix} F & C_a \end{bmatrix}, \quad L_e := \begin{bmatrix} L \\ -B_a \end{bmatrix}$$

and where A_e, B_e, C_e are as in (12.5). □

12.5 Closed-Loop Transfer Matrix

Recall that the closed-loop transfer matrix from w to z is a linear fractional transformation $\mathcal{F}_\ell(G, K)$ and that K stabilizes G if and only if K stabilizes G_{22} . Elimination of the signals u and y in Figure 12.3 leads to Figure 12.4 for a suitable transfer matrix T . Thus all closed-loop transfer matrices are representable as in Figure 12.4.

$$z = \mathcal{F}_\ell(G, K)w = \mathcal{F}_\ell(G, \mathcal{F}_\ell(J, Q))w = \mathcal{F}_\ell(T, Q)w. \quad (12.7)$$

It remains to give a realization of T .

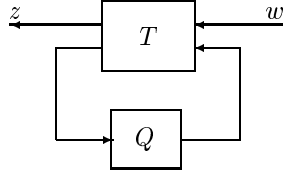


Figure 12.4: Closed loop system

Theorem 12.16 *Let F and L be such that $A + BF$ and $A + LC$ are stable. Then the set of all closed-loop transfer matrices from w to z achievable by an internally stabilizing proper controller is equal to*

$$\mathcal{F}_\ell(T, Q) = \{T_{11} + T_{12}QT_{21} : Q \in \mathcal{RH}_\infty, I + D_{22}Q(\infty) \text{ invertible}\}$$

where T is given by

$$T = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} = \left[\begin{array}{cc|cc} A + B_2F & -B_2F & B_1 & B_2 \\ 0 & A + LC_2 & B_1 + LD_{21} & 0 \\ \hline C_1 + D_{12}F & -D_{12}F & D_{11} & D_{12} \\ 0 & C_2 & D_{21} & 0 \end{array} \right].$$

Proof. This is straightforward by using the state space star product formula and follows from some tedious algebra. Hence it is left for the readers to verify. $\square$

An important point to note is that the closed-loop transfer matrix is simply an affine function of the controller parameter matrix Q since $T_{22} = 0$.

12.6 Youla Parameterization via Coprime Factorization*

In this section, all stabilizing controller parameterization will be derived using the conventional coprime factorization approach. Readers should be familiar with the results presented in Section 5.4 of Chapter 5 before proceeding further.

Theorem 12.17 *Let $G_{22} = NM^{-1} = \tilde{M}^{-1}\tilde{N}$ be the rcf and lcf of G_{22} over $\mathcal{RH}_\infty$, respectively. Then the set of all proper controllers achieving internal stability is parameterized either by*

$$K = (U_0 + MQ_r)(V_0 + NQ_r)^{-1}, \quad \det(I + V_0^{-1}NQ_r)(\infty) \neq 0 \quad (12.8)$$

for $Q_r \in \mathcal{RH}_\infty$ or by

$$K = (\tilde{V}_0 + Q_l\tilde{N})^{-1}(\tilde{U}_0 + Q_l\tilde{M}), \quad \det(I + Q_l\tilde{N}\tilde{V}_0^{-1})(\infty) \neq 0 \quad (12.9)$$

for $Q_l \in \mathcal{RH}_\infty$ where $U_0, V_0, \tilde{U}_0, \tilde{V}_0 \in \mathcal{RH}_\infty$ satisfy the Bezout identities:

$$\tilde{V}_0 M - \tilde{U}_0 N = I, \quad \tilde{M} V_0 - \tilde{N} U_0 = I.$$

Moreover, if $U_0, V_0, \tilde{U}_0$, and $\tilde{V}_0$ are chosen such that $U_0 V_0^{-1} = \tilde{V}_0^{-1} \tilde{U}_0$, i.e.,

$$\begin{bmatrix} \tilde{V}_0 & -\tilde{U}_0 \\ -\tilde{N} & \tilde{M} \end{bmatrix} \begin{bmatrix} M & U_0 \\ N & V_0 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}.$$

Then

$$\begin{aligned} K &= (U_0 + M Q_y)(V_0 + N Q_y)^{-1} \\ &= (\tilde{V}_0 + Q_y \tilde{N})^{-1} (\tilde{U}_0 + Q_y \tilde{M}) \\ &= \mathcal{F}_\ell(J_y, Q_y) \end{aligned} \quad (12.10)$$

where

$$J_y := \begin{bmatrix} U_0 V_0^{-1} & \tilde{V}_0^{-1} \\ V_0^{-1} & -V_0^{-1} N \end{bmatrix} \quad (12.11)$$

and where Q_y ranges over $\mathcal{RH}_\infty$ such that $(I + V_0^{-1} N Q_y)(\infty)$ is invertible

Proof. We shall prove the parameterization given in (12.8) first. Assume that K has the form indicated, and define

$$U := U_0 + M Q_r, \quad V := V_0 + N Q_r.$$

Then

$$\tilde{M} V - \tilde{N} U = \tilde{M}(V_0 + N Q_r) - \tilde{N}(U_0 + M Q_r) = \tilde{M} V_0 - \tilde{N} U_0 + (\tilde{M} N - \tilde{N} M) Q_r = I.$$

Thus K achieves internal stability by Lemma 5.10.

Conversely, suppose K is proper and achieves internal stability. Introduce an rcf of K over $\mathcal{RH}_\infty$ as $K = UV^{-1}$. Then by Lemma 5.10, $Z := \tilde{M} V - \tilde{N} U$ is invertible in $\mathcal{RH}_\infty$. Define Q_r by the equation

$$U_0 + M Q_r = U Z^{-1}, \quad (12.12)$$

so

$$Q_r = M^{-1}(U Z^{-1} - U_0).$$

Then using the Bezout identity, we have

$$\begin{aligned} V_0 + N Q_r &= V_0 + N M^{-1}(U Z^{-1} - U_0) \\ &= V_0 + \tilde{M}^{-1} \tilde{N}(U Z^{-1} - U_0) \\ &= \tilde{M}^{-1}(\tilde{M} V_0 - \tilde{N} U_0 + \tilde{N} U Z^{-1}) \\ &= \tilde{M}^{-1}(I + \tilde{N} U Z^{-1}) \\ &= \tilde{M}^{-1}(Z + \tilde{N} U) Z^{-1} \\ &= \tilde{M}^{-1} \tilde{M} V Z^{-1} \\ &= V Z^{-1}. \end{aligned} \quad (12.13)$$

Thus,

$$\begin{aligned} K &= UV^{-1} \\ &= (U_0 + MQ_r)(V_0 + NQ_r)^{-1}. \end{aligned}$$

To see that Q_r belongs to $\mathcal{RH}_\infty$, observe first from (12.12) and then from (12.13) that both MQ_r and NQ_r belong to $\mathcal{RH}_\infty$. Then

$$Q_r = (\tilde{V}_0 M - \tilde{U}_0 N)Q_r = \tilde{V}_0(MQ_r) - \tilde{U}_0(NQ_r) \in \mathcal{RH}_\infty.$$

Finally, since V and Z evaluated at $s = \infty$ are both invertible, so is $V_0 + NQ_r$ from (12.13), hence so is $I + V_0^{-1}NQ_r$.

Similarly, the parameterization given in (12.9) can be obtained.

To show that the controller can be written in the form of equation (12.10), note that

$$(U_0 + MQ_y)(V_0 + NQ_y)^{-1} = U_0 V_0^{-1} + (M - U_0 V_0^{-1} N)Q_y(I + V_0^{-1} NQ_y)^{-1} V_0^{-1}$$

and that $U_0 V_0^{-1} = \tilde{V}_0^{-1} \tilde{U}_0$. We have

$$(M - U_0 V_0^{-1} N) = (M - \tilde{V}_0^{-1} \tilde{U}_0 N) = \tilde{V}_0^{-1} (\tilde{V}_0 M - \tilde{U}_0 N) = \tilde{V}_0^{-1}$$

and

$$K = U_0 V_0^{-1} + \tilde{V}_0^{-1} Q_y (I + V_0^{-1} NQ_y)^{-1} V_0^{-1}. \quad (12.14)$$

□

Corollary 12.18 *Given an admissible controller K with coprime factorizations $K = UV^{-1} = \tilde{V}^{-1}\tilde{U}$, the free parameter $Q_y \in \mathcal{RH}_\infty$ in Youla parameterization is given by*

$$Q_y = M^{-1}(UZ^{-1} - U_0)$$

where

$$Z := \tilde{M}V - \tilde{N}U.$$

Next, we shall establish the precise relationship between the above all stabilizing controller parameterization and the parameterization obtained in the previous sections via LFT framework.

Theorem 12.19 *Let the doubly coprime factorizations of G_{22} be chosen as*

$$\left[\begin{array}{cc|cc} M & U_0 & A + B_2 F & B_2 & -L \\ N & V_0 & F & I & 0 \\ C_2 + D_{22} F & & D_{22} & I & \end{array} \right]$$

$$\begin{bmatrix} \tilde{V}_0 & -\tilde{U}_0 \\ -\tilde{N} & \tilde{M} \end{bmatrix} = \left[\begin{array}{c|cc} A + LC_2 & -(B_2 + LD_{22}) & L \\ \hline F & I & 0 \\ C_2 & -D_{22} & I \end{array} \right]$$

where F and L are chosen such that $A + B_2F$ and $A + LC_2$ are both stable.

Then J_y can be computed as

$$J_y = \left[\begin{array}{cc|cc} A + B_2F + LC_2 + LD_{22}F & -L & B_2 + LD_{22} & \\ \hline F & 0 & I & \\ -(C_2 + D_{22}F) & I & -D_{22} & \end{array} \right].$$

Proof. This follows from some tedious algebra. $\square$

Remark 12.11 Note that J_y is exactly the same as the J in Theorem 12.8 and that $K_0 := U_0V_0^{-1}$ is an observer-based stabilizing controller with

$$K_0 := \left[\begin{array}{c|c} A + B_2F + LC_2 + LD_{22}F & -L \\ \hline F & 0 \end{array} \right].$$

♡

12.7 Notes and References

The special problems FI, DF, FC, and OE were first introduced in Doyle, Glover, Khar-gonekar, and Francis [1989] for solving the $\mathcal{H}_\infty$ problem, and they have been since used in many other papers for different problems. The new derivation of all stabilizing controllers was reported in Lu, Zhou, and Doyle [1991]. The paper by Moore *et al* [1990] contains some other related interesting results. The conventional Youla parameterization can be found in Youla *et al* [1976], Desoer *et al* [1980], Doyle [1984], Vidyasagar [1985], and Francis [1987]. The parameterization of all two-degree-of-freedom stabilizing controllers is given in Youla and Bongiorno [1985] and Vidyasagar [1985].